

Muneeb Abbasi

Principal AI Engineer — LLMs, RAG & agentic systems · 15+ years · ex-Microsoft

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Summary

Principal Software Engineer with 15+ years delivering enterprise-scale software, AI platforms, cloud-native systems, and workflow automation. Hands-on expertise in LLMs, RAG, AI agents, Azure, .NET Core, Python, React, microservices, and data platforms. Proven ability to lead technical discovery and solution architecture with customers, then own delivery end to end — from design through production adoption — with measurable business outcomes.

Technical expertise

AI & applied intelligence	Generative AI, LLMs, RAG, AI agents, agentic workflows, semantic & vector search, prompt engineering, OpenAI / Azure OpenAI, LangGraph, OCR, scikit-learn
Cloud & infrastructure	Microsoft Azure, Azure Functions, Event Hub, Data Factory, Azure AI Services, Docker, Kubernetes, GitHub Actions, Azure DevOps, event-driven architecture
Software engineering	Python, C#, .NET Core, TypeScript, JavaScript, Java, SQL, React, Angular, FastAPI, microservices, REST APIs
Data & storage	PostgreSQL, SQL Server, Azure SQL, MongoDB, vector databases, data pipelines, telemetry analytics

Professional experience

Principal AI Engineer — Global E-commerce & Retail

Bellevue, WA · Feb 2025 – Present

Enterprise AI platform leveraging LLMs, RAG, AI agents, semantic search, and machine learning to modernize inventory, supply chain, procurement, logistics, sales, and customer operations.

- Designed and delivered enterprise AI platforms with LLMs, RAG, and AI agents, reducing manual reconciliation by 90%
- Built scalable cloud-native applications using Python, FastAPI, React, PostgreSQL, and microservices
- Led customer-facing technical discovery, architecture workshops, and solution demonstrations through to production adoption
- Established monitoring and continuous optimization to improve AI performance and operational reliability

Senior Software Engineer — Microsoft

Redmond, WA · Oct 2014 – Jan 2025

Led technical discovery, solution architecture, and delivery of cloud-native automation and AI-powered platforms for Microsoft's global network infrastructure, owning solutions from requirements through production while remaining a hands-on engineer.

Network Infrastructure Copilot (NIC)

- Built an enterprise AI copilot (LLMs, RAG, semantic search, telemetry) for natural language access to KPIs, incidents, and operational systems
- Increased engineering productivity by 35% and reduced incident investigation time across global infrastructure

Enterprise AI knowledge & orchestration platform

- Designed RAG and vector-search architecture unifying enterprise knowledge across engineering systems and documentation
- Reduced engineering research and decision-making time by 60%; improved onboarding and cross-team collaboration

Network device automation system

- Designed cloud-native microservices (.NET Core, Azure Functions, event-driven architecture) orchestrating global deployment workflows
- Reduced manual device scheduling by 60% with 99.9% platform availability

Business process optimization platform

- Automated enterprise workflows with orchestration, telemetry, operational intelligence, and AI-driven recommendations
- Improved business process efficiency by 40% and increased customer adoption

Education

- Master of Business Information Systems (MBS) — University College Cork
- Bachelor of Engineering in Information Technology (B.Eng.) — Hamdard University